

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVPE604C - OPEN CHANNEL FLOW

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Open channel flow is characterized by**
 - A) Flow under pressure
 - B) Flow with free surface exposed to atmosphere
 - C) Flow through closed conduit
 - D) Flow due to pump action
- 2 Flow in an open channel depends mainly on**
 - A) Pressure difference
 - B) Density of fluid
 - C) Gravity
 - D) Temperature
- 3 Which instrument is commonly used for point velocity measurement in open channels?**
 - A) Venturimeter
 - B) Orificemeter
 - C) Current meter
 - D) Piezometer
- 4 Velocity distribution in open channel flow is maximum**
 - A) At the bed
 - B) At the free surface
 - C) Slightly below the free surface
 - D) At mid-depth
- 5 Manning's roughness coefficient depends on**
 - A) Channel slope
 - B) Channel material and surface condition
 - C) Flow velocity
 - D) Discharge

- 6 **A hydraulically most efficient channel section has**
 - A) Maximum wetted perimeter
 - B) Minimum hydraulic radius
 - C) Maximum discharge for given area
 - D) Minimum discharge
- 7 **Specific energy in open channel flow is defined as**
 - A) Energy per unit weight at channel bed
 - B) Energy per unit area
 - C) Energy per unit mass
 - D) Total energy loss
- 8 **Flow at critical depth is characterized by**
 - A) Maximum velocity
 - B) Minimum specific energy
 - C) Maximum discharge
 - D) Zero energy
- 9 **Hydraulic jump is an example of**
 - A) Uniform flow
 - B) Gradually varied flow
 - C) Rapidly varied flow
 - D) Laminar flow
- 10 **Cipolletti weir is a type of**
 - A) Triangular notch
 - B) Broad crested weir
 - C) Trapezoidal weir
 - D) Rectangular weir
- 11 **Backwater curves are formed due to**
 - A) Sudden increase in slope
 - B) Obstruction or rise in downstream water level
 - C) Decrease in discharge
 - D) Uniform channel section
- 12 **The conjugate depths in a hydraulic jump are related using**
 - A) Continuity equation
 - B) Energy equation
 - C) Momentum equation
 - D) Bernoulli's equation

Q2. Solve the following.

- A) Write the differences between pipe flow and open channel flow? 6
- B) Explain the energy equation as applied to open channel flow. 6

Q3. Solve the following.

- A) Derive the conditions for hydraulically most efficient rectangular section. 6
- B) State Chezy's formula for uniform flow and explain the significance of each term. 6

Q4. Solve Any Two of the following.

- A) Define specific energy and explain in detail with neat sketch of specific energy curve? 6
- B) Explain critical, sub-critical and super-critical flow with respect to specific energy. 6

- C) A 12 m wide rectangular channel carries a discharge of $25 \text{ m}^3/\text{sec}$ with a depth of 3m. Find the width to which the channel should be contracted to get critical flow at the contracted section. prepare similar numerical 6

Q5. Solve Any Two of the following.

- A) Explain with neat sketch the concept of hydraulic jump. 6
- B) Define gradually varied flow (GVF). What are the few typical examples of GVF? Explain the two basic assumptions in the analysis of GVF? 6
- C) In a rectangular channel, the depth of flow before a hydraulic jump is 0.8 m and the velocity is 8 m/s. Determine the conjugate depth after the jump. 6

Q6. Solve Any Two of the following.

- A) Define weir. Classify weirs based on shape and crest. 6
- B) Explain discharge over a rectangular weir including velocity of approach. 6
- C) During an experiment in a laboratory, 0.05 m^3 of water flowing over a right angled notch was collected in one minute. If the head of the sill is 50 mm, calculate the coefficient of discharge of the notch. 6

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